



Bala Venkatesh

Lead Data Scientist
AI / ML Engineer | GenAI
Architect

◆ SCB Best Performer 2024 ◆
8+ Years Experience

CONTACT

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[LinkedIn Profile](#)
[GitHub Portfolio](#)
[Personal Website](#)
[Medium Articles](#)

AI & GENAI

- ◆ Large Language Models (LLMs)
- ◆ Generative AI Architecture
- ◆ RAG Pipeline Design
- ◆ LangChain / LangGraph
- ◆ Multi-Agent Orchestration
- ◆ CrewAI / AutoGen / Prefect
- ◆ Prompt Engineering & Tuning
- ◆ AI System Design & Strategy
- ◆ Vector DB (Pinecone, FAISS)
- ◆ OpenAI / Azure OpenAI APIs

ML & ENGINEERING

- ◆ Python — Advanced
- ◆ Deep Learning / CNNs
- ◆ Computer Vision (OpenCV)
- ◆ NLP & Text Processing
- ◆ Model Fine-Tuning & MLOps
- ◆ FastAPI / Django REST
- ◆ TensorFlow / PyTorch

CLOUD & DEVOPS

- ◆ Microsoft Azure (Certified)
- ◆ Azure DevOps (ADO)
- ◆ AWS EC2 / S3 / Lambda
- ◆ Databricks
- ◆ Docker / CI-CD Pipelines
- ◆ Git / GitHub

PROFESSIONAL SUMMARY

Award-winning **Senior Data Scientist and AI/ML Engineer** with **8+ years** delivering production-grade GenAI and ML systems across global **banking, healthcare, and energy**. Recognised as **SCB Bank Best Performer 2024** for pioneering AI-driven security automation. Expert in **LLM architecture, RAG pipelines, multi-agent systems**, and scalable cloud deployments on **Azure and AWS**. **MSc in Data Science** (Teesside University, UK). Published author with **25+ technical articles** across Medium, KDnuggets, and TopCoder. Open-source contributor and strategic AI thought leader.

KEY ACHIEVEMENTS

- ▶ **SCB Best Performer 2024** — Recognised for engineering **RAG-based security automation** that transformed SCB's DevOps security posture across global pipelines
- ▶ **GenAI Platform Architect** — Sole architect of **Phoenix CLM platform**, automating contract management for a **top-5 global bank** using LLMs + Prefect orchestration
- ▶ **Social Impact AI** — Built **Crop Doctor app** helping Indian farmers detect plant diseases via deep learning; published **25+ technical articles** on Medium, KDnuggets & TopCoder
- ▶ **Security AI Innovation** — Reduced **Azure DevOps misconfiguration risk** at enterprise scale using **prompt-engineered LLM remediation** with exact file/line precision

WORK EXPERIENCE

Phoenix — AI Contract Lifecycle Management (CLM) *Oct 2025 – Present*

Standard Chartered Bank · India (Hybrid)

Sr. ML Engineer | GenAI Architect

- ▶ Architected **end-to-end GenAI platform** for SCB's legal contract portfolio — automated metadata extraction, classification, clause analysis and pipeline orchestration
- ▶ Built **LLM-powered extraction engine** eliminating manual tagging across thousands of **high-value legal contracts**
- ▶ Designed **AI classifiers** to auto-identify contract types (master, sub, standalone, order forms, variations) at enterprise scale
- ▶ Implemented **clause similarity scoring engine** to flag template deviations — **significantly reducing legal risk review time**

TIP — Threat Identification & Prevention *Sep 2023 – Sep 2025*

Standard Chartered Bank · India (Hybrid)

ML Engineer | AI Security Automation Lead

- ▶ Led full design and delivery of **RAG-based AI system** automating misconfiguration detection across **Azure DevOps pipelines** globally
- ▶ Engineered **Vector DB store** for TIP security controls; integrated **LangChain** for intelligent semantic retrieval against live pipeline configs
- ▶ Crafted **prompt-tuned LLM responses** citing exact file/line numbers with step-by-step remediation — eliminating ambiguity for engineers
- ▶ Automated full DevOps validation loop; dramatically improved security posture → **awarded SCB Bank Best Performer 2024**

Crop Doctor — AI Crop Disease Detection Application *Mar 2022 – Mar 2023*

Bighat PVT LTD · India (Remote, Contract)

Lead Data Scientist

- ▶ Directed **end-to-end ML development** of mobile app enabling farmers to detect plant diseases via **deep learning image analysis**
- ▶ Designed training pipeline, curated datasets, **fine-tuned CNN models**; built continuous model update loop for real-world accuracy

Medosys — AI Healthcare Documentation Platform *Oct 2020 – Jul 2021*

AIBOTS Sdn Bhd · Johor Bahru, Malaysia

Data Scientist

LEADERSHIP & MANAGEMENT

- Team Leadership & Mentoring
- AI Roadmap & Strategy
- Stakeholder Management
- Agile / Scrum
- Technical Architecture
- Cross-functional Delivery
- Jira / Azure Boards

CERTIFICATIONS

- Microsoft Azure — Multiple
- Udemy — Adv. Data Science

EDUCATION

- MSc Applied Data Science
Teesside University, UK
Sep 2021 – Sep 2022
- BE Computer Science
Sri Ram Engineering College
Jul 2011 – Apr 2015

LANGUAGES

- Tamil — Native
- English — Professional

- Trained NLP model for clinical entity extraction from physician–patient conversations — auto-generating structured medical notes
- Led data science team; managed delivery cadence, sprint alignment, and client communication across the full project lifecycle

Phraze — Voice-to-EMR Clinical AI Scribe Dec 2018 – Jan 2020

Spritle Software Pvt Ltd · Chennai, India
Data Scientist

- Built NLP-powered voice documentation pipeline enabling physicians to auto-generate Electronic Medical Records — freeing doctors to focus on patients
- Designed AI scribe solution using voice interface and clinical NLP to improve quality and quantity of data collected
- Gathered requirements, conducted ML research and implementation; drove code quality through peer reviews

Tools Monitoring — Prison Security Object Detection Jul 2018 – Nov 2018

Spritle Software Pvt Ltd · Chennai, India
Data Scientist

- Built automated object detection system to identify and log tools carried by workers entering and exiting Singapore prison facilities
- System verified tool inventory on exit — ensuring no tools were left inside, improving prison safety compliance

Moddo — B2B E-Commerce Platform (Footwear Industry) Jan 2018 – Jul 2018

Spritle Software Pvt Ltd · Chennai, India
Senior Software Developer

- Developed features for Moddo — the world's leading B2B e-commerce platform for footwear, component and accessory professionals
- Built modules for trade offers, company reports, trade fair calendars, fashion trends, news and subscription services

Mack Lite & Time Transport System Oct 2016 – Dec 2017

Solverminds Technologies Pvt Ltd · Chennai, India
Software Developer → Junior Software Developer

- Mack Lite: Implemented maritime shipping records system to manage safety precautions, ship inspection reports, and shore communication workflows
- Time Transport: Automated transport operations — vehicle and employee tracking using narrow-search concepts; engaged with Product Owner for requirements
- Designed and developed highly interactive, responsive web applications for both platforms

CORE COMPETENCIES

 GenAI & LLM - LLM Pipeline Architecture - RAG System Design - LangChain / LangGraph - Multi-Agent Orchestration - Prompt Engineering - Vector DB Integration	 ML & Data Science - Python (Advanced) - Deep Learning / CNNs - Computer Vision - NLP & Text Mining - MLOps & Model Serving - TensorFlow / PyTorch	 Leadership - AI Roadmap & Vision - Team Leadership - Stakeholder Management - Agile / Scrum - Technical Architecture - Azure / AWS Cloud
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PUBLICATIONS & COMMUNITY

25+ Technical Articles · Amazon Book Author · Open-Source Contributor

- Medium: TensorFlow Object Detection, ML Setup, Blockchain & Web3.0, ChatBot (DialogFlow), Deep Learning image rotation
- KDnuggets & TopCoder: Computer Vision with TensorFlow.js · Data Science vs Statistics · What is OpenCV?
- Amazon Book & GitHub: Contributing author — Data Science & Business Intelligence (Amazon B08BPB2KKK) · github.com/balavenkatesh3322